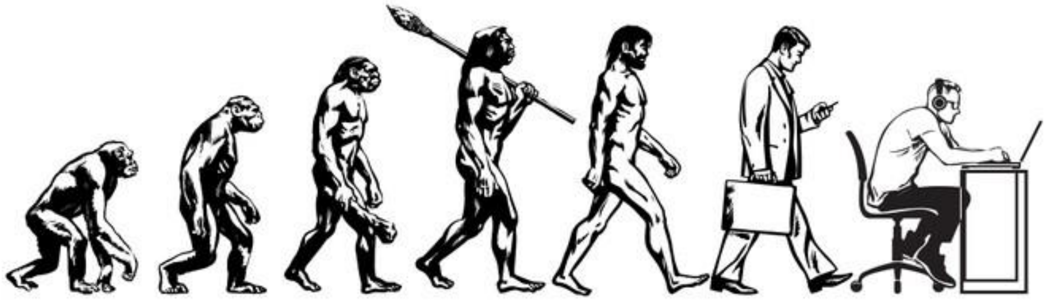
Genetic Algorithm

Gunner Cook-Dumas



Why use Genetic Algorithm

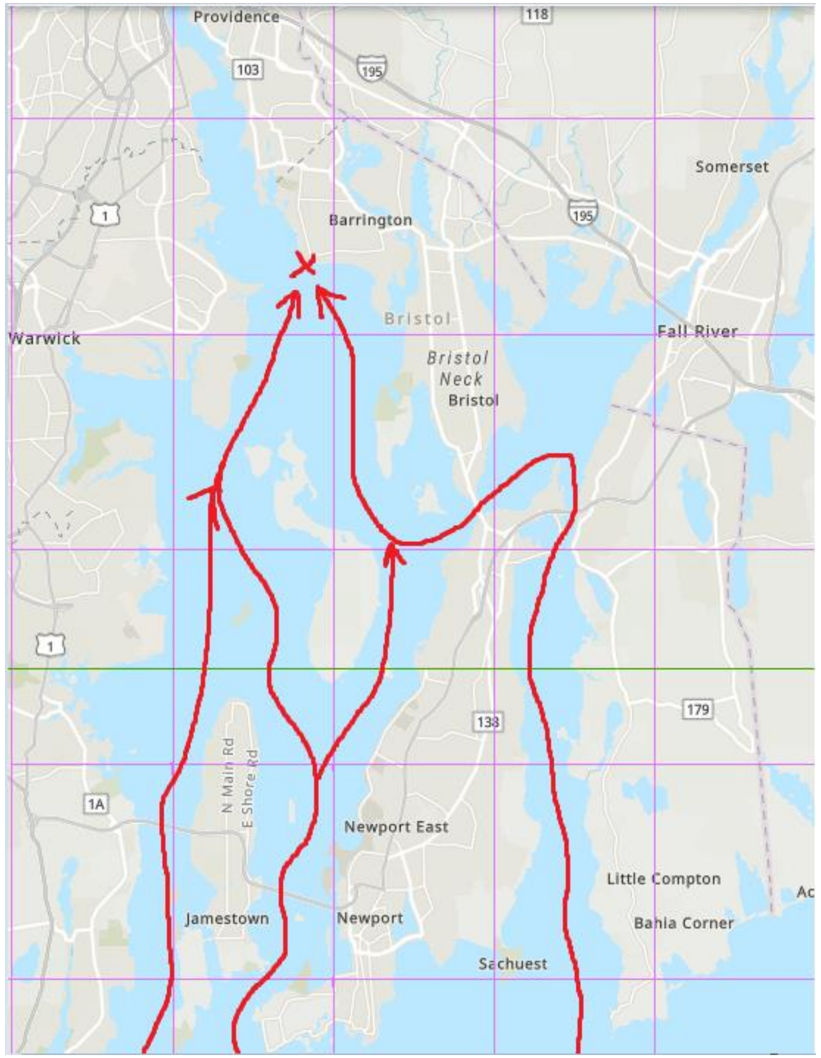
- Explore multiple best options
- Don’t need to explicitly program it
- Can generate multiple optimal results
- Can use the results to optimize other AI frameworks

Other work

- Finding optimal carry weight YouTube
- Optimization of GA medium
- Maze solving
- Word solving
- Jump King by CodeBullet

Problem

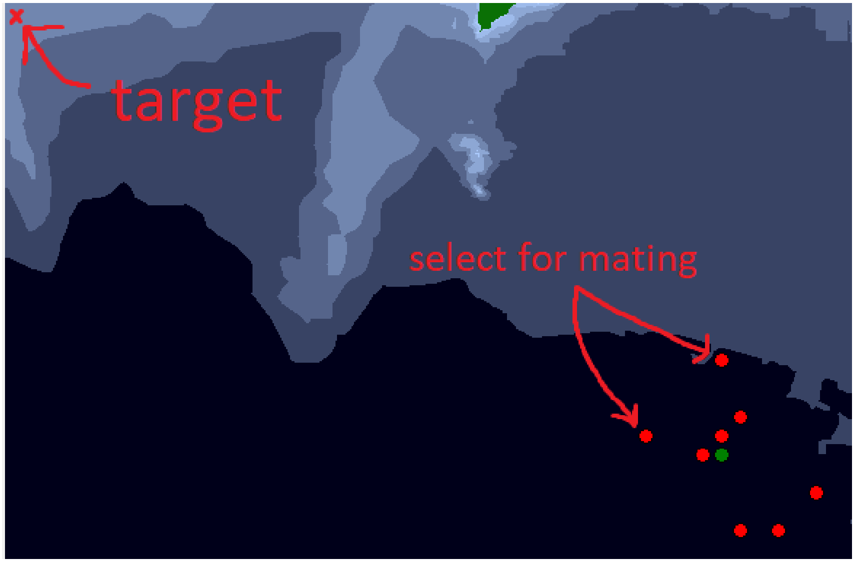
- I need to search a coast and navigate obstacles
- There are multiple solutions
- How do I avoid detection?
- Which one is fastest, stealthiest?



What have I done

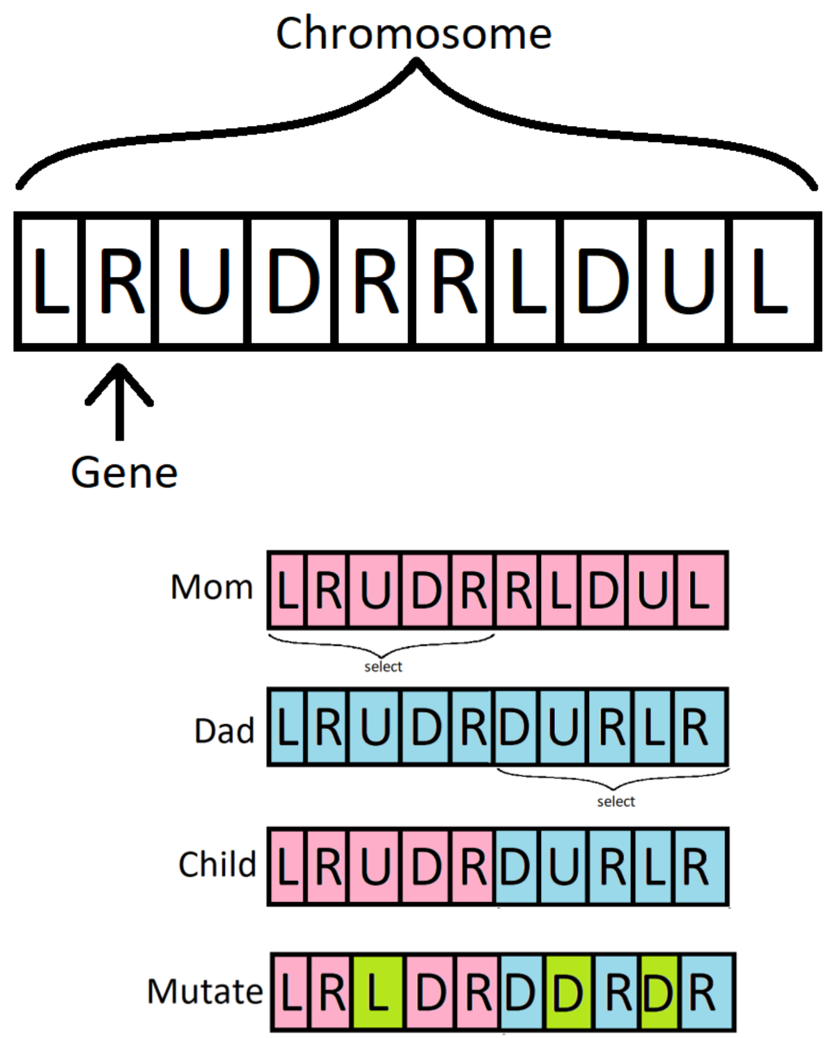
- Built a Genetic algorithm with agent-based modeling
- Agents will try to find an optimal solution to water based environment
- Fitness is calculated by using a Manhattan distance to target

$$Distance = |x_1 - x_2| + |y_1 - y_2|$$



How I did it

- Each agent contains a chromosome
- Each chromosome is a list of genes
- Find the two best parents
- Mating is half-n-half
- Mutate the child



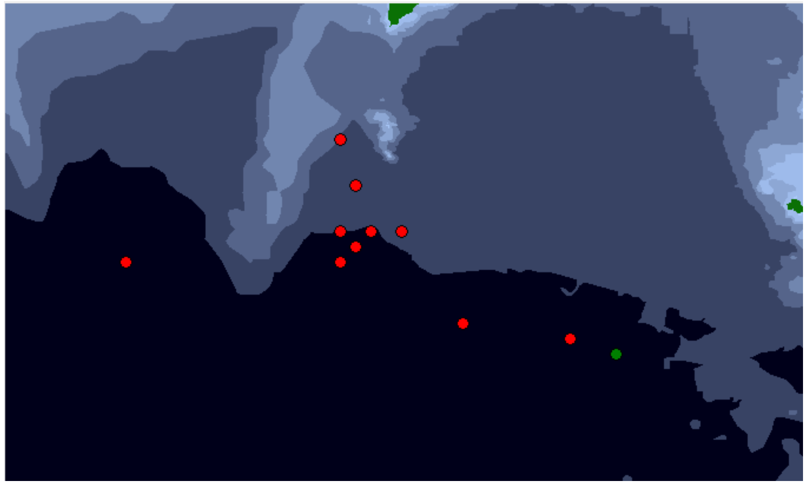
Working so far?

- The GA correctly pass on genes
- The GA "learn" to get closer to the target through the generations

Target is top left corner

Generation 1

Target is top left corner



Generation 20

What are some problems

- Limits to how many generations I can make
- Chromosomes need more diversity
  - Child makes same mistakes as parents
- Better path finding as lots of backtracking and circles

What I plan too do

- Fitness function needs to be tailored
  - Add more parameters
  - Include more parents for diversity
- Crossover function
  - The half-n-half is not diverse enough
  - Chromosomes need more randomness
- Fix wall collision